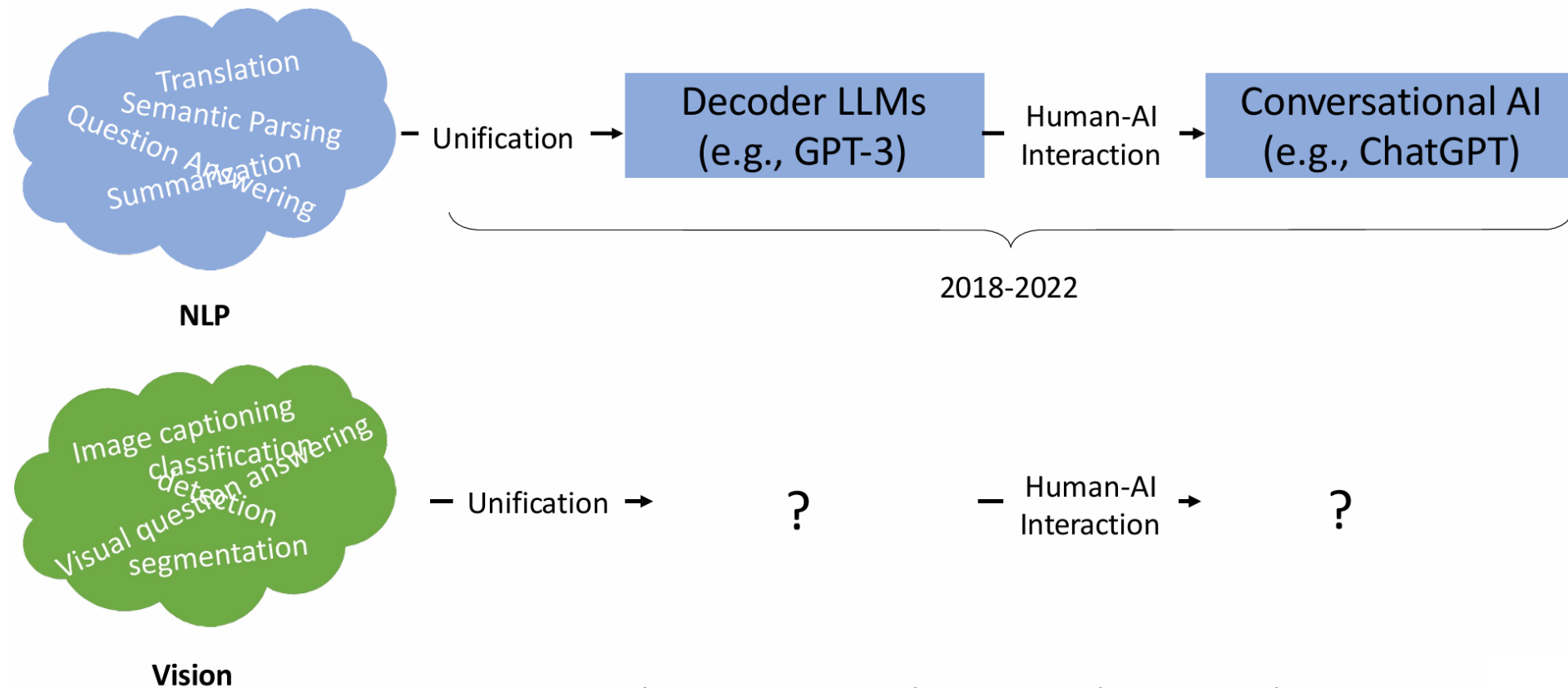


Multi-modal Foundation Models

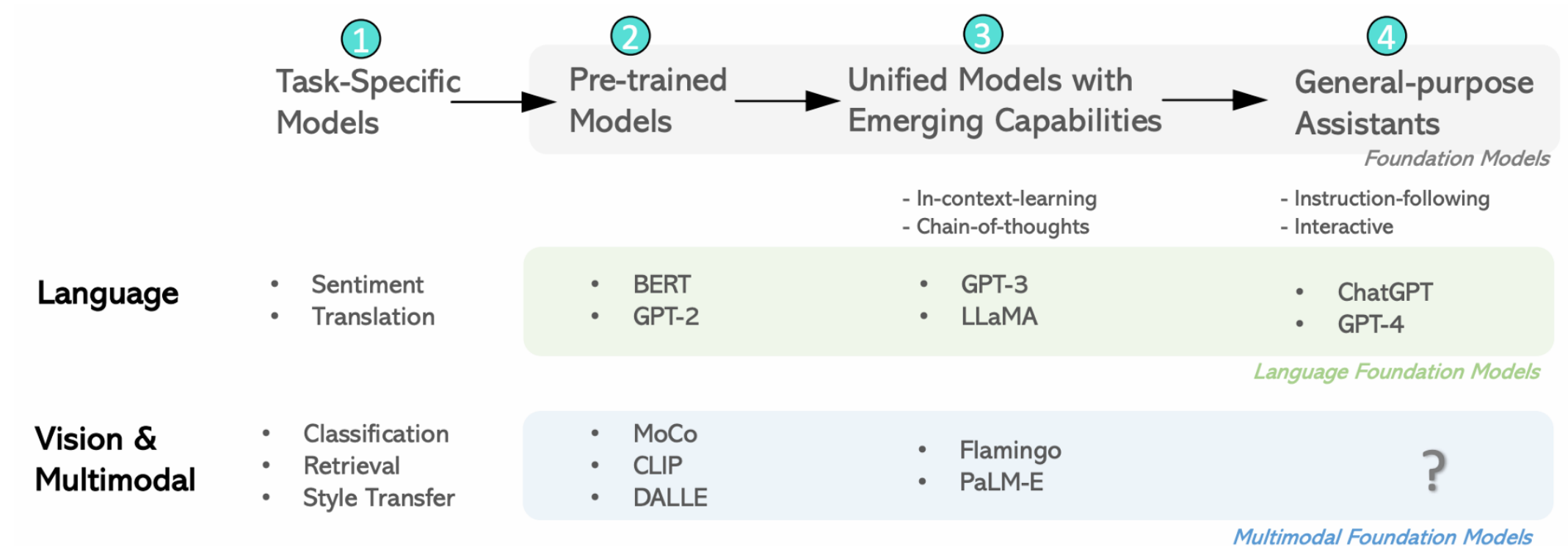


A Lesson from LLMs



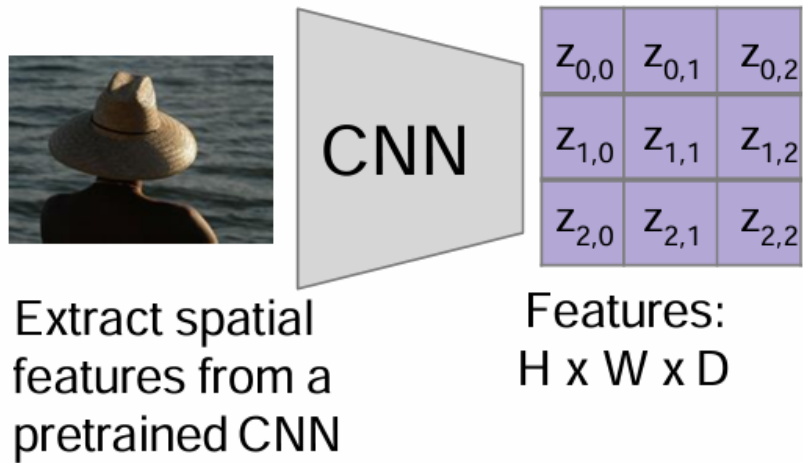
Jianwei Yang's CVPR 2023 Tutorial: From Specialist to Generalist

A Lesson from LLMs



Transformer in Vision

- Image Captioning using Transformers



Transformer in Vision

Input: Image I

Output: Sequence $\mathbf{y} = y_1, y_2, \dots, y_T$

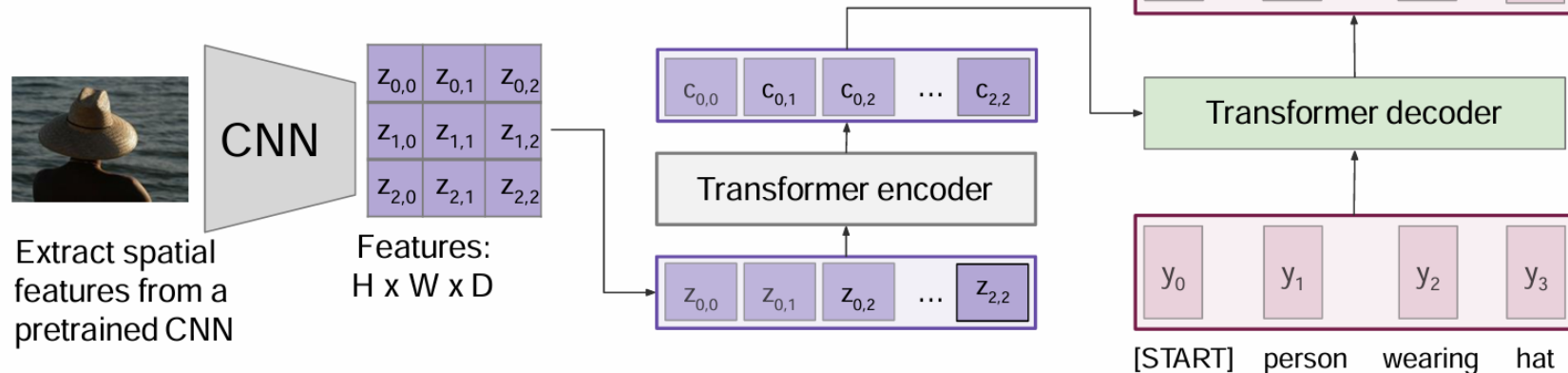
Decoder: $y_t = T_D(y_{0:t-1}, c)$

Where $T_D(\cdot)$ is the transformer decoder

Encoder: $c = T_w(\mathbf{z})$

where $\mathbf{z}$ is spatial CNN features

$T_w(\cdot)$ is the transformer encoder



Transformer in Vision

Perhaps we don't need convolutions at all?

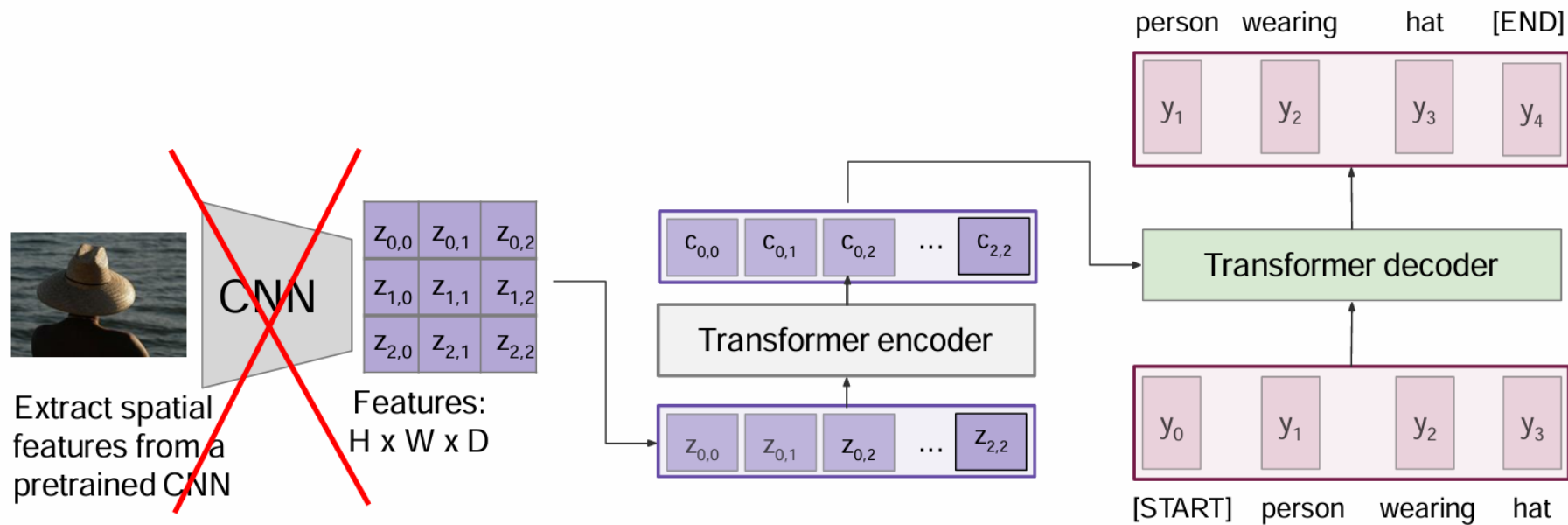
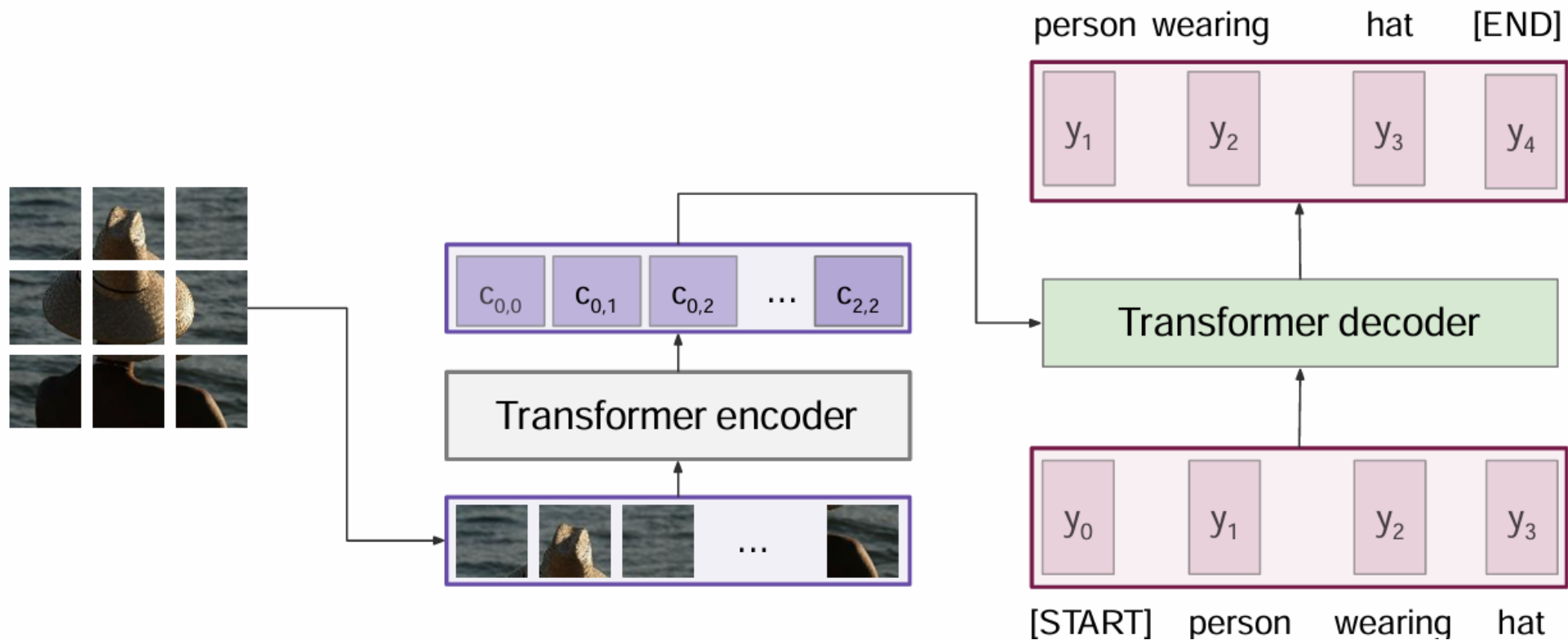
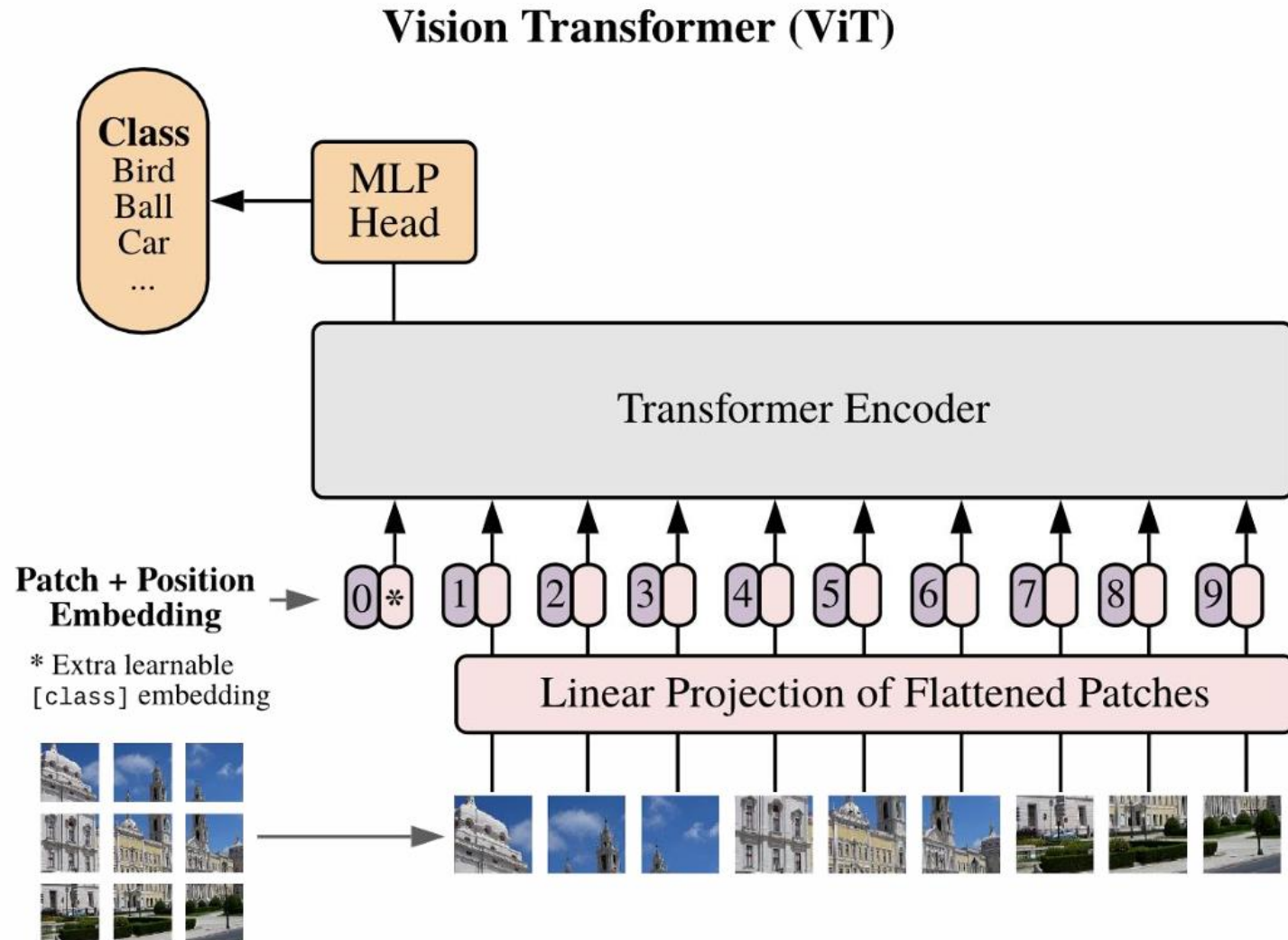


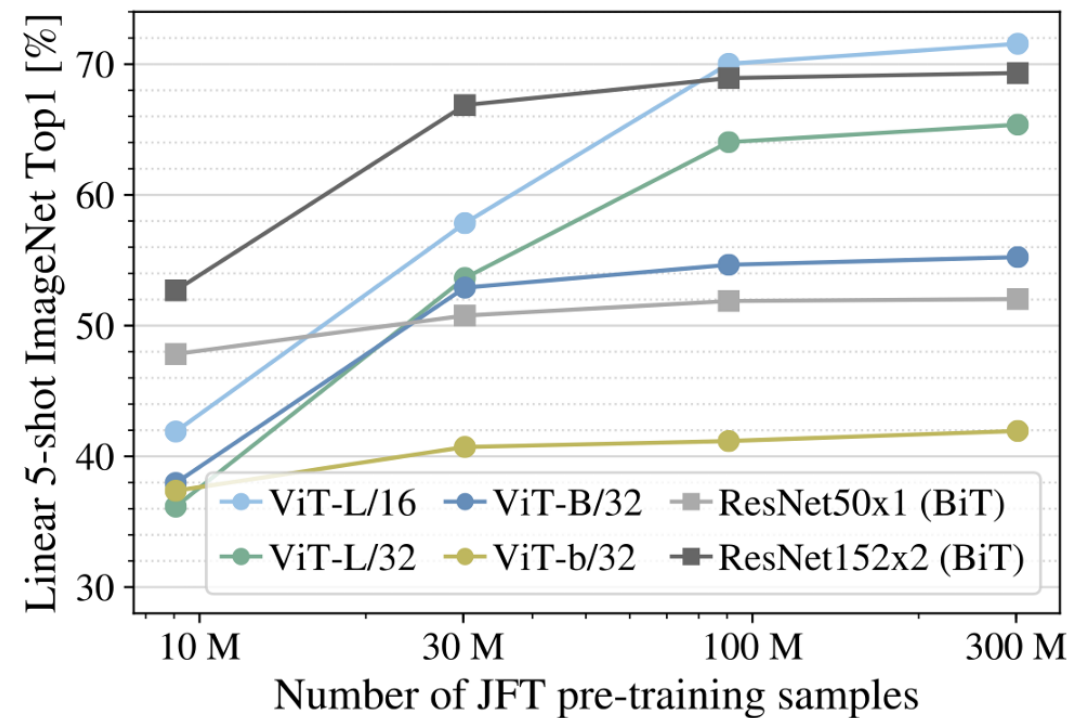
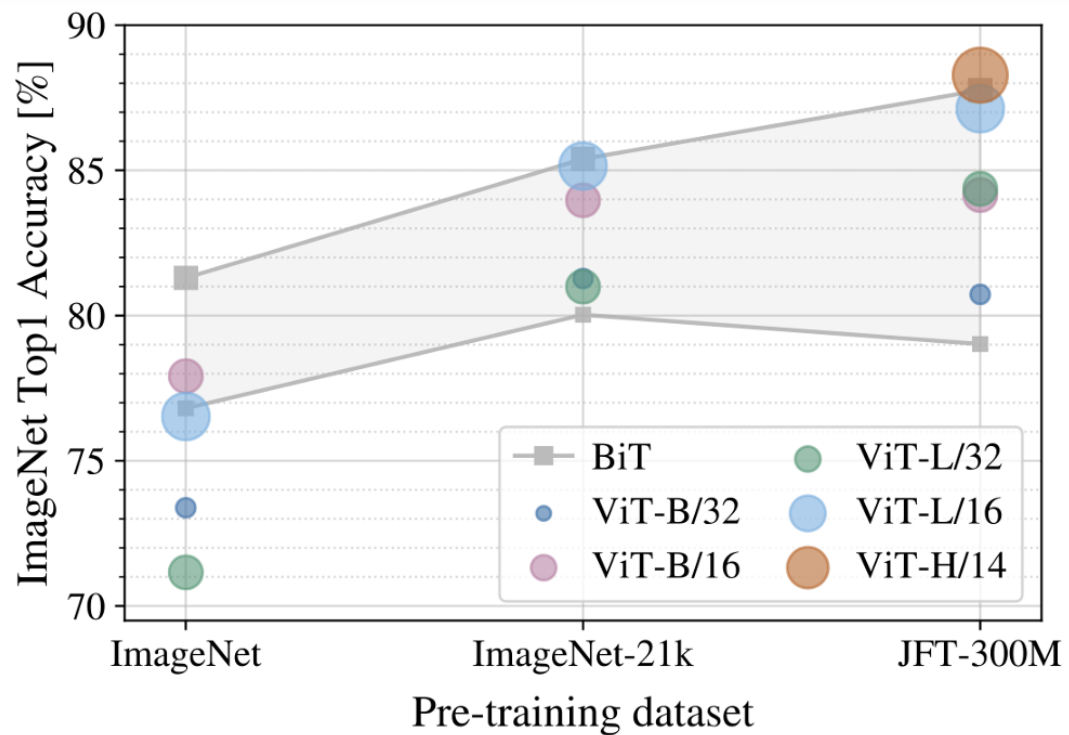
Image Captioning using **ONLY** transformers



ViT as Image Encoder Architecture



Vision Transformers vs. ResNets



Vision Transformers vs. ResNets

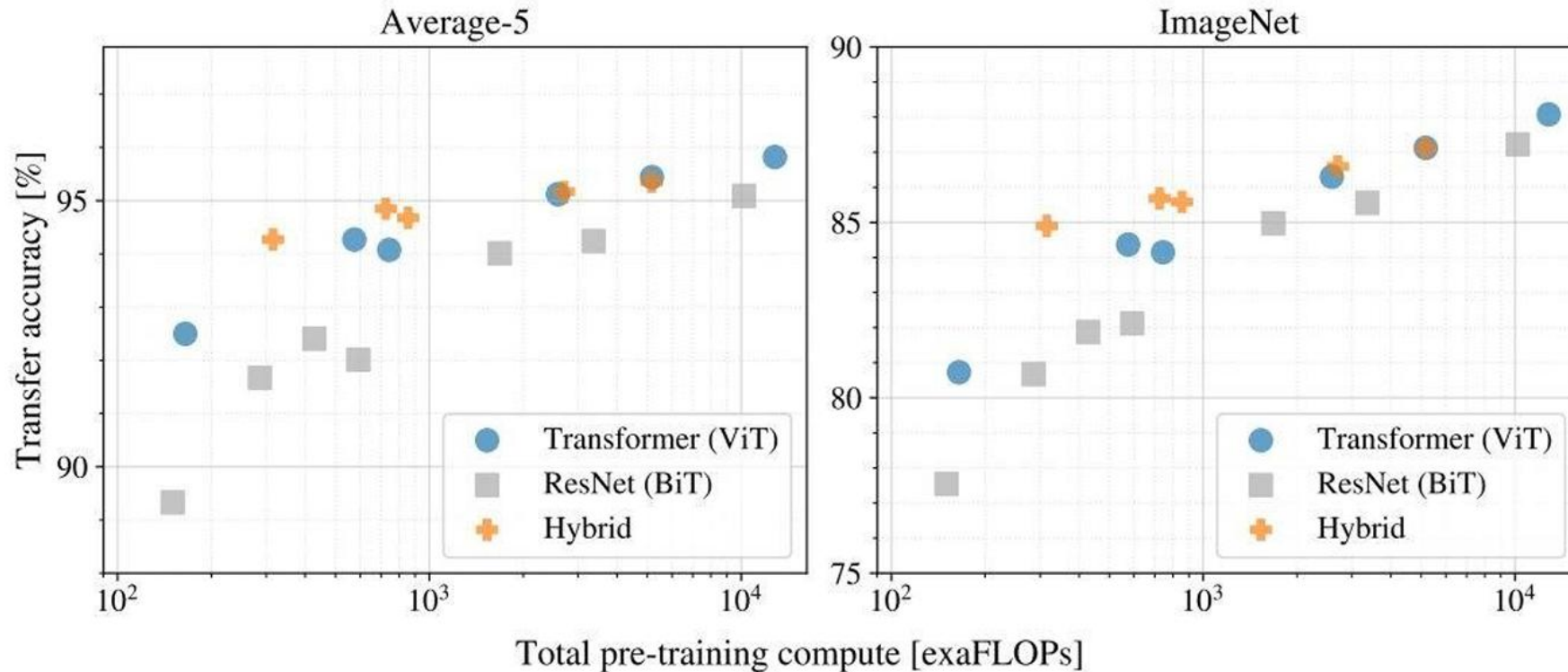
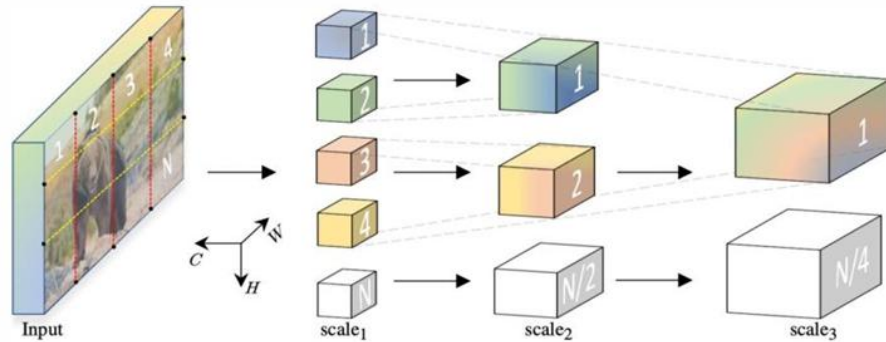
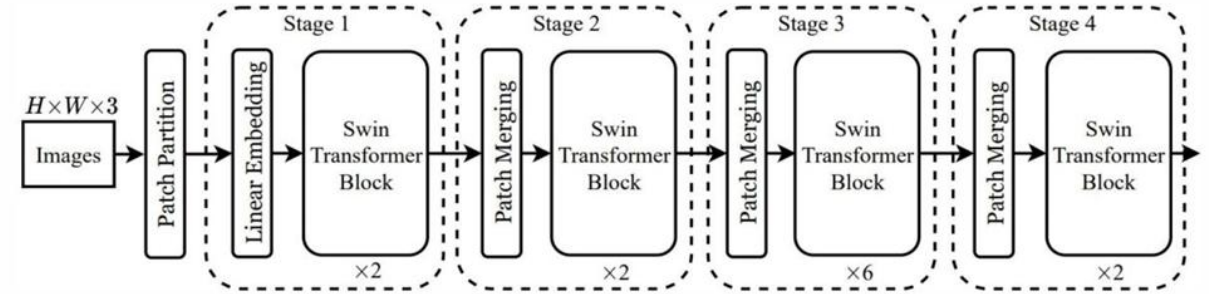


Figure 5: Performance versus cost for different architectures: Vision Transformers, ResNets, and hybrids. Vision Transformers generally outperform ResNets with the same computational budget. Hybrids improve upon pure Transformers for smaller model sizes, but the gap vanishes for larger models.

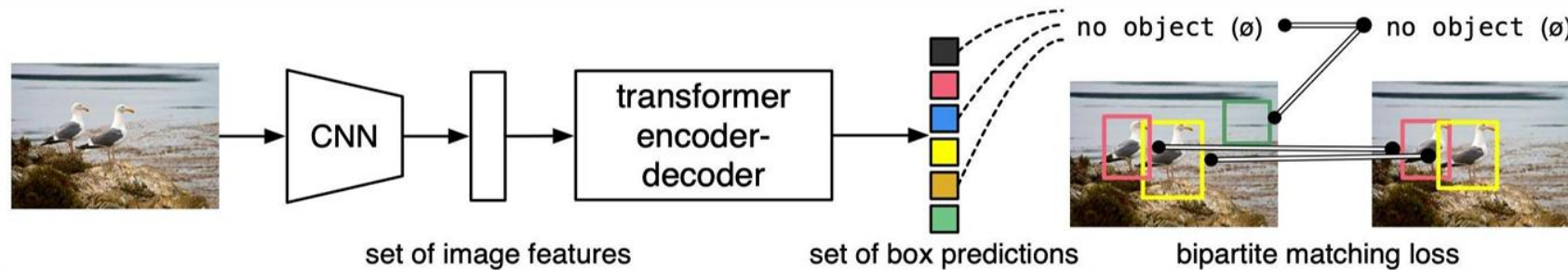
Vision Transformers



Fan et al, "Multiscale Vision Transformers", ICCV 2021

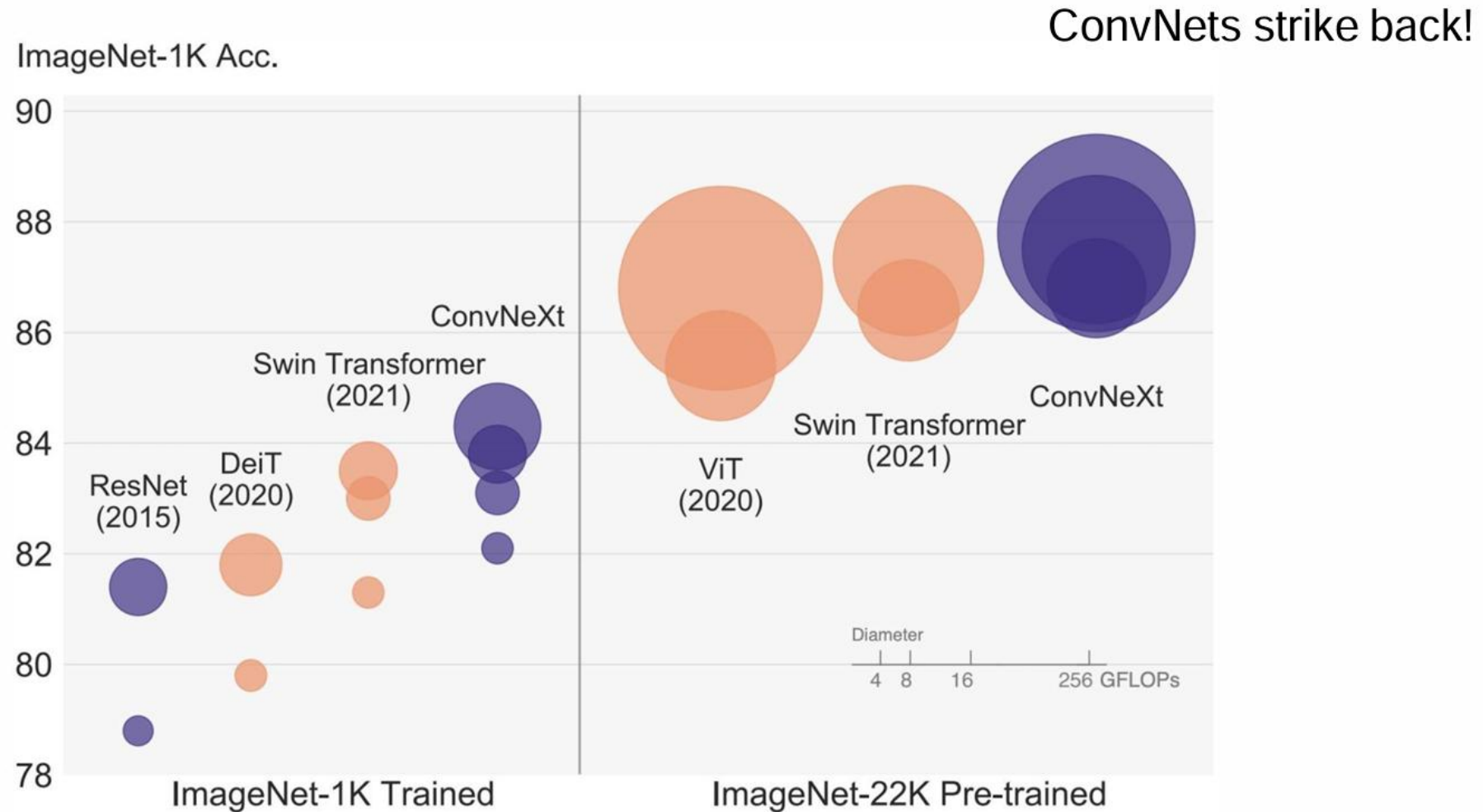


Liu et al, "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows", CVPR 2021



Carion et al, "End-to-End Object Detection with Transformers", ECCV 2020

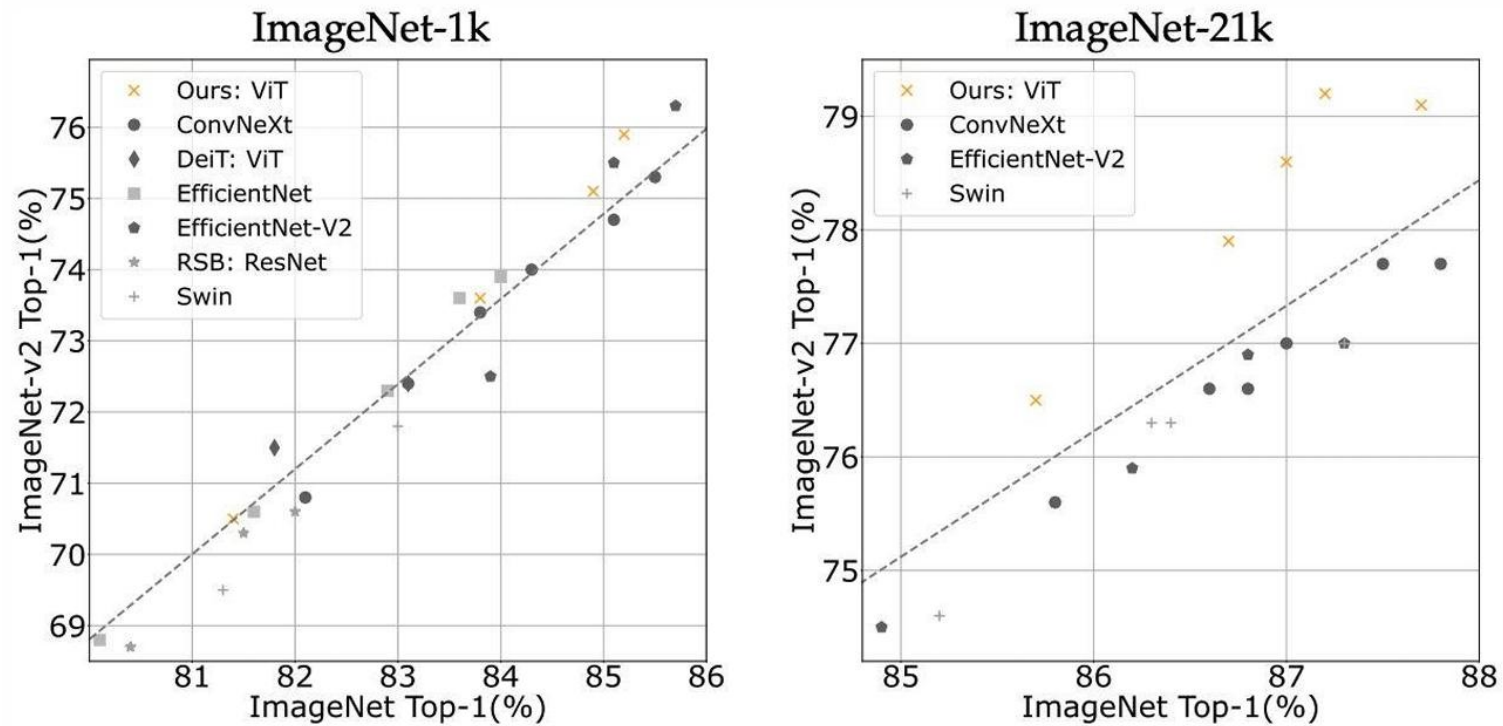
Vision Transformers



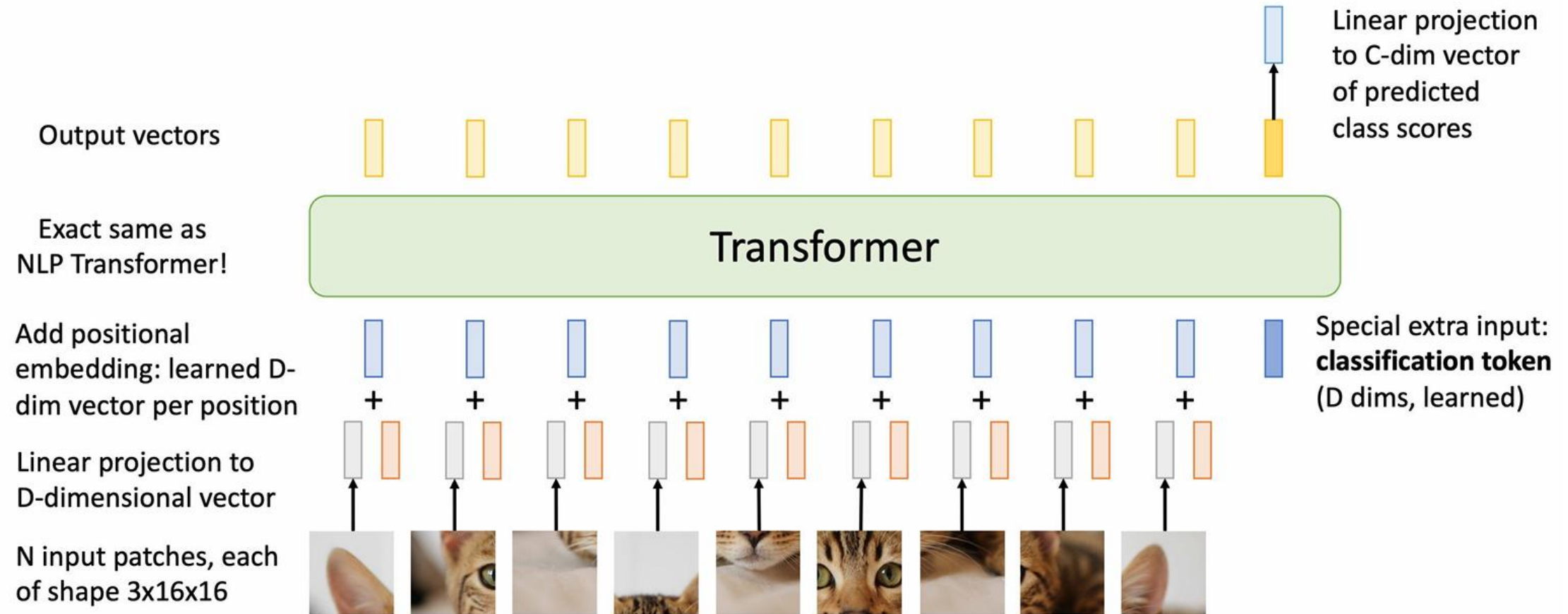
Revenge of the ViT

DeiT III: Revenge of the ViT

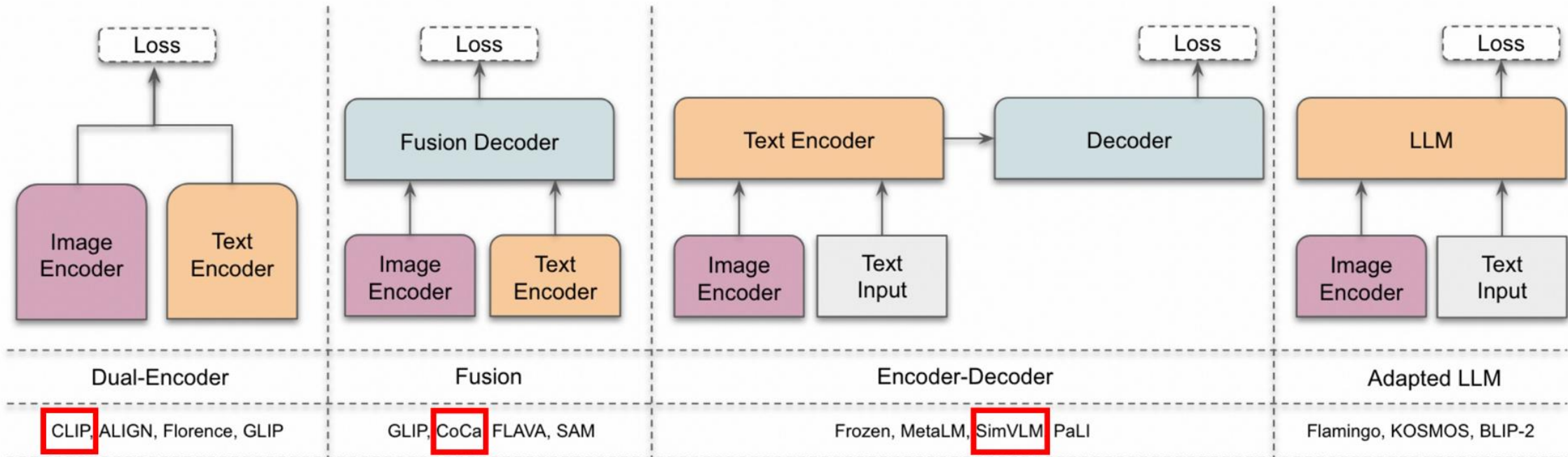
Hugo Touvron^{*,†} Matthieu Cord[†] Hervé Jégou^{*}



Vision Language Model



Different Architecture



Survey paper

Multimodal Foundation Models: From Specialists to General-Purpose Assistants

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Lijuan Wang, Jianfeng Gao**

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* Core Contribution ♠ Project Lead